

2.1 Convolutional Neural Network

We consider a 1D CNN with L layers, where each layer i has kernel size K_i and stride S_i . Let X_0 denote the input, and X_i denote the activations after the i -th convolutional layer.

(a) Effective Stride ES_i

The **effective stride** ES_i represents the step size in X_0 corresponding to one step in X_i . It can be defined recursively as:

$$ES_i = ES_{i-1} \times S_i,$$

with base case:

$$ES_1 = S_1.$$

Hence, the general formula becomes:

$$ES_i = \prod_{j=1}^i S_j.$$

(b) Receptive Field R_i

The **receptive field** R_i of a neuron in X_i denotes the number of pixels in X_0 that affect its value. It can be defined recursively as:

$$R_i = R_{i-1} + (K_i - 1) \times ES_{i-1},$$

with base case:

$$R_1 = K_1.$$

Thus, the general closed-form expression is:

$$R_i = 1 + \sum_{j=1}^i (K_j - 1) \prod_{m=1}^{j-1} S_m.$$

(c) Example Verification

For $K_1 = K_2 = 3$, $S_1 = S_2 = 2$:

$$ES_1 = 2,$$

$$ES_2 = 2 \times 2 = 4,$$

$$R_1 = 3,$$

$$R_2 = 3 + (3 - 1) \times 2 = 7.$$

This matches the figure where $ES_1 = 2$, $ES_2 = 4$, $R_1 = 3$, and $R_2 = 7$.